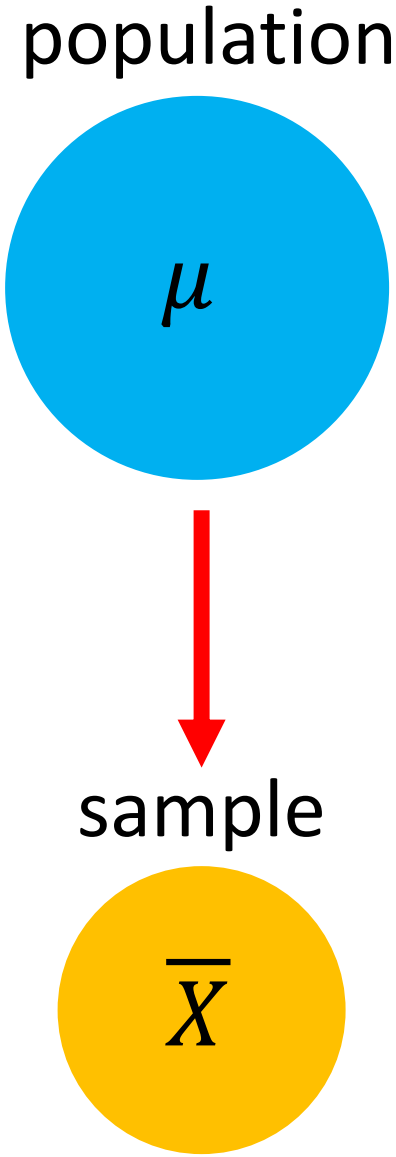
Unit 07: Confidence Interval

Dr. Md Mutasim Billah

95% CONFIDENCE INTERVAL FOR μ WHEN σ IS KNOWN

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think beyond the possible℠



Inferential Statistics

Using data to draw conclusions about a population and measuring the reliability of the conclusions.

Point Estimate

Estimate μ with $\overline{X}$

Confidence Interval

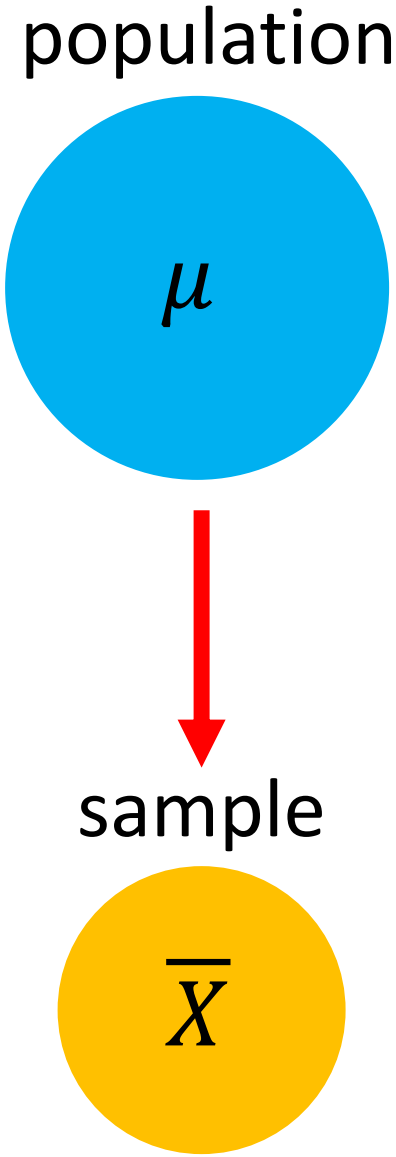
Calculate an interval of plausible values for μ

$\overline{X} \pm$ error margin = (A, B)

Hypothesis Test

Use the sample data to try to answer a question about μ

Example: Does $\mu = 8$? Yes or no?



Consider this Situation:

We are about to take a random sample from a population.

1

Suppose the population is normally distributed or the sample size is large ($n \geq 30$).

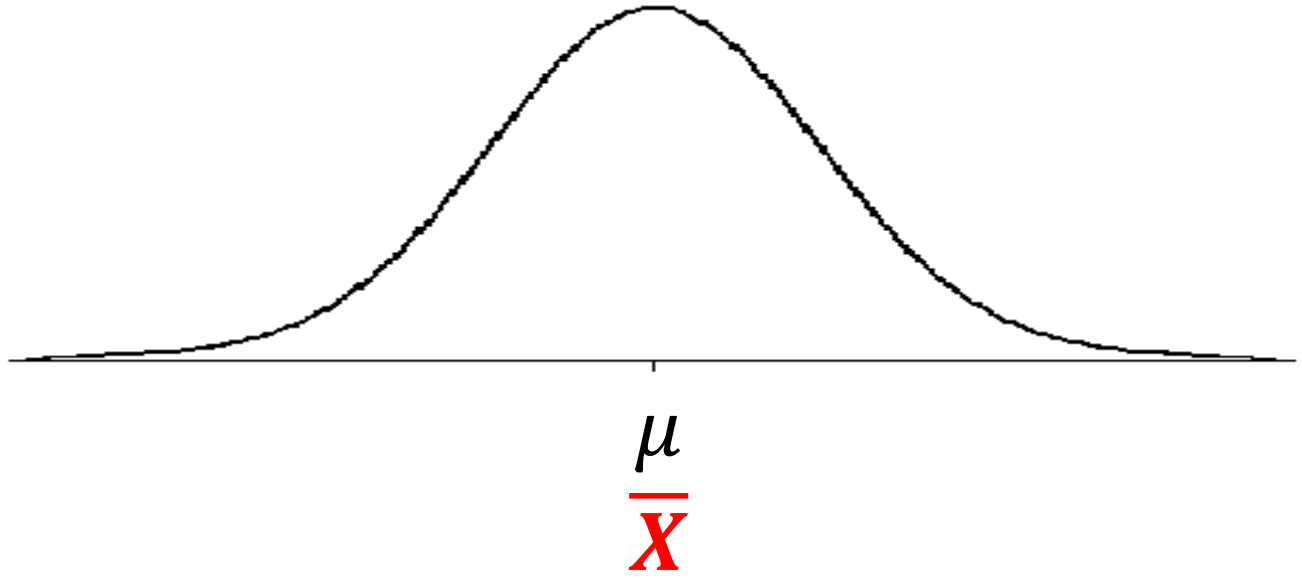
2

Suppose σ is known (unrealistic)

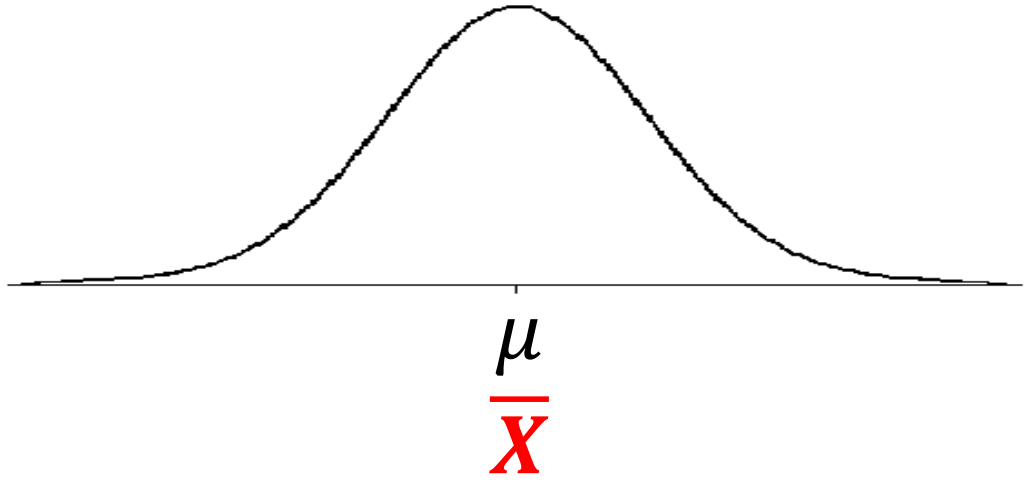
Thanks to the Central Limit Theorem, we actually know quite a bit about the sample mean we are about to observe.

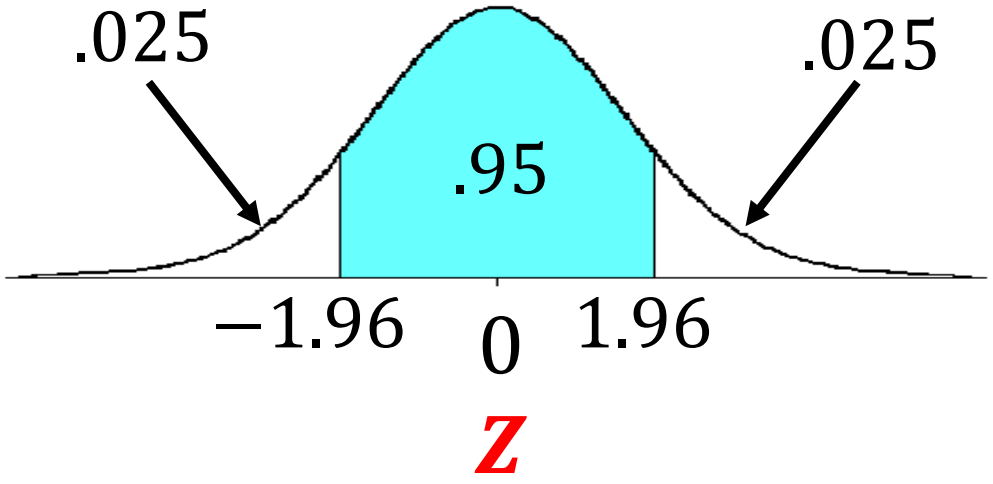
Since the population is normally distributed or the sample size is large ($n \geq 30$), the sample mean we are about to observe will come from a normal distribution with a mean of $E(\overline{X}) = \mu$ and a standard deviation of $SD(\overline{X}) = \frac{\sigma}{\sqrt{n}}$

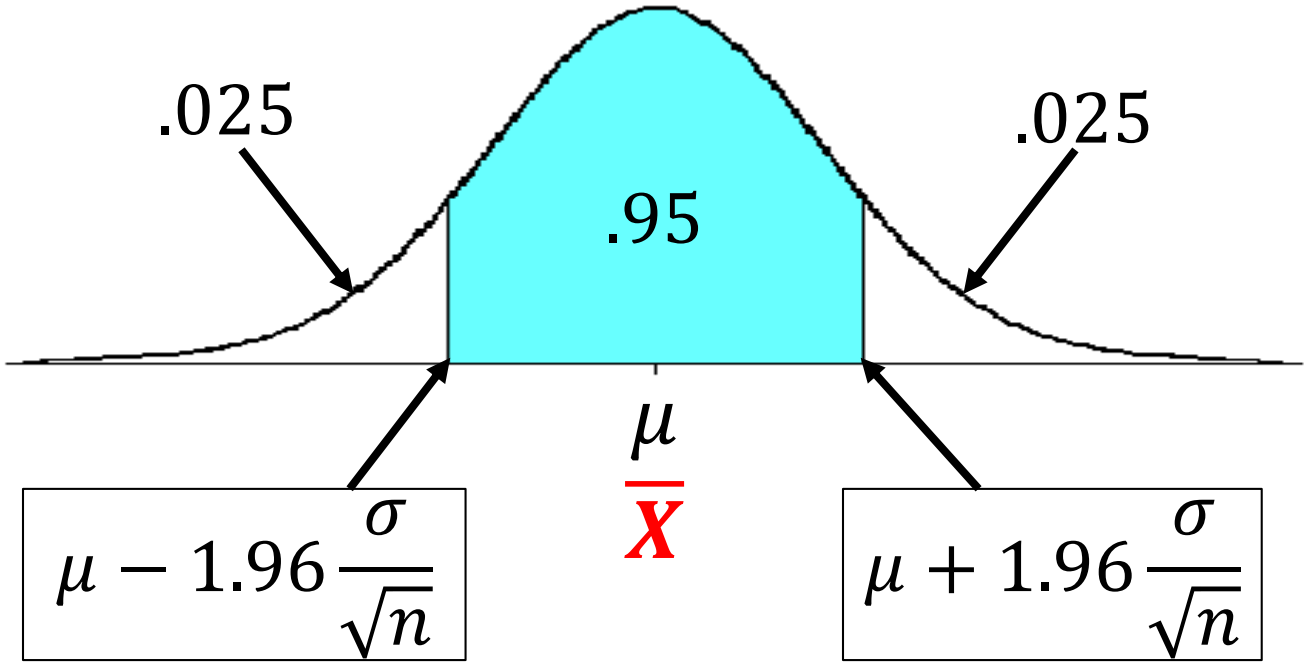
$$\overline{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$



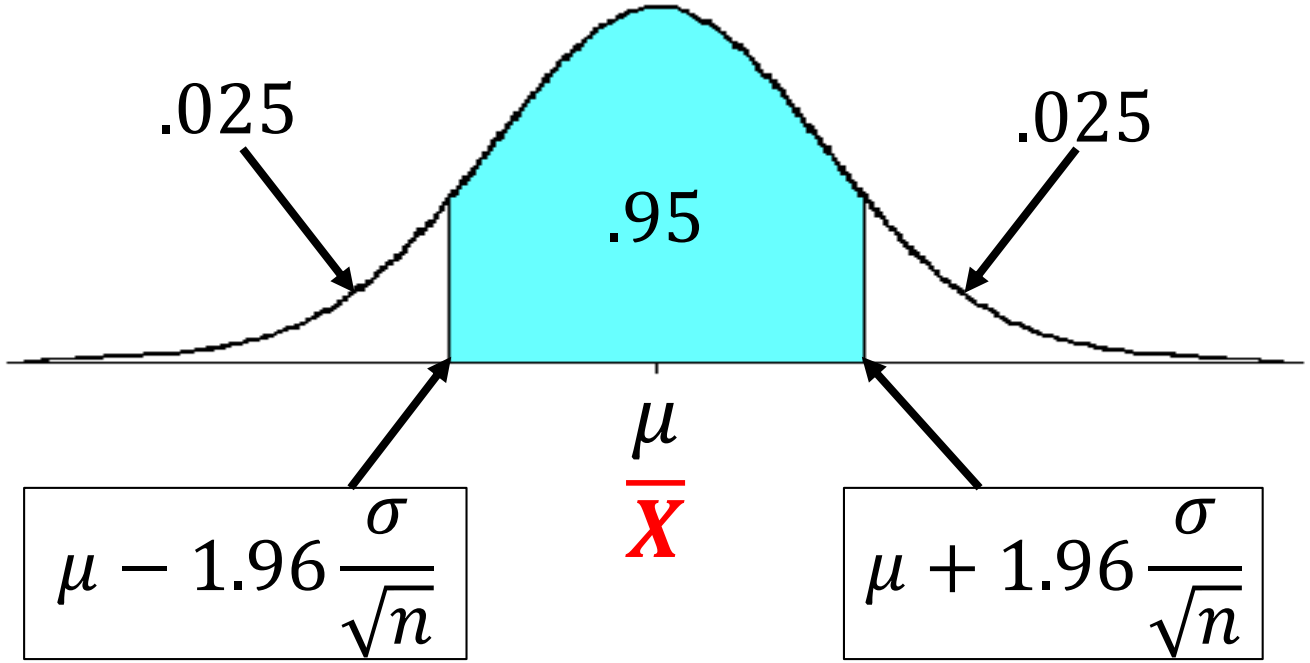
$$\overline{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$







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$$.95 = P\left(\mu - 1.96\frac{\sigma}{\sqrt{n}} < \overline{X} < \mu + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

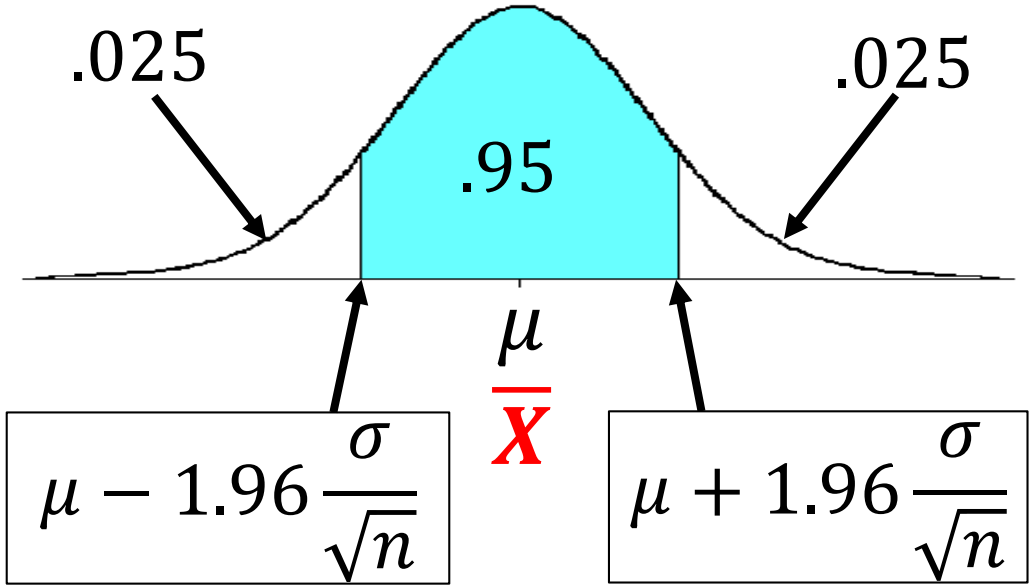
$$.95 = P\left(\mu - 1.96\frac{\sigma}{\sqrt{n}} < \overline{X} < \mu + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

$$.95 = P\left(-1.96\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right)$$

$$.95 = P\left(-\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < -\mu < -\overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

$$.95 = P\left(\overline{X} + 1.96\frac{\sigma}{\sqrt{n}} > \mu > \overline{X} - 1.96\frac{\sigma}{\sqrt{n}}\right)$$

$$.95 = P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

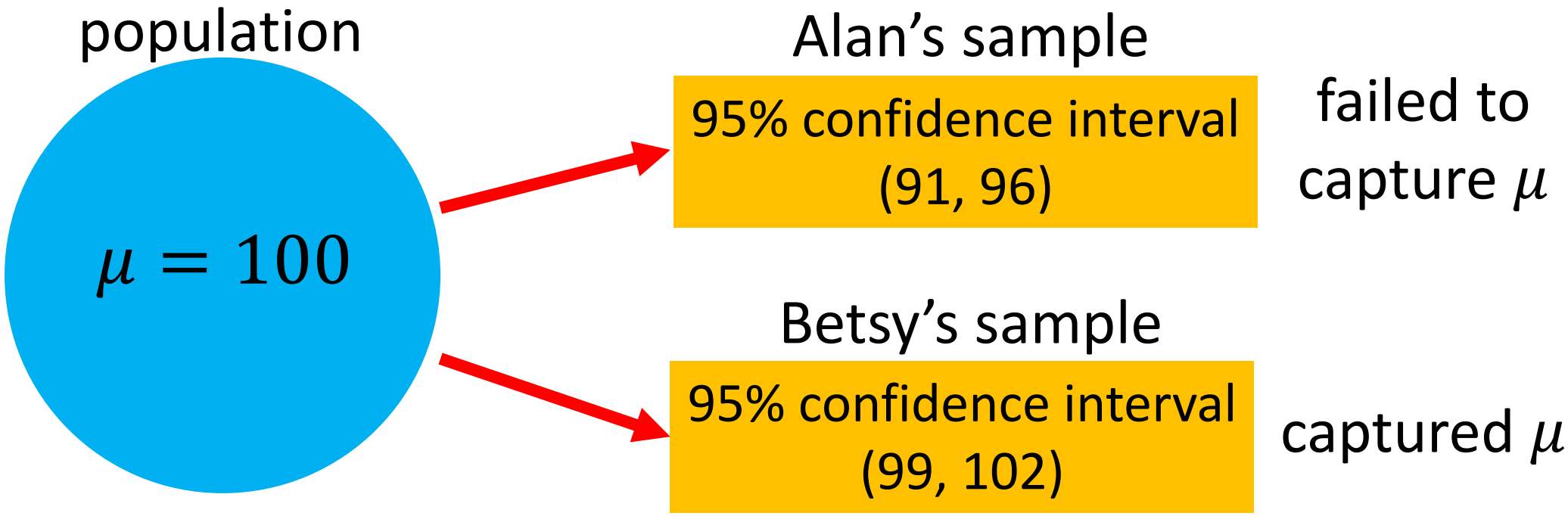


$$.95 = P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

Conclusion:

95% of the time we take a random sample from a normally distributed population (or have a large sample size), the interval $\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$ will "capture" μ.

95% of the time we take a random sample from a normally distributed population (or have a large sample size), the interval $\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$ will "capture" μ.



95% confidence interval for μ when σ is known

$$\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) \rightarrow \overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Suppose we are interested in μ = the population mean weight among all Michigan Tech students.

$n = 30$

$\overline{X} = 150$ pounds

$\sigma = 25$ (known)

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{30}}\right)$$

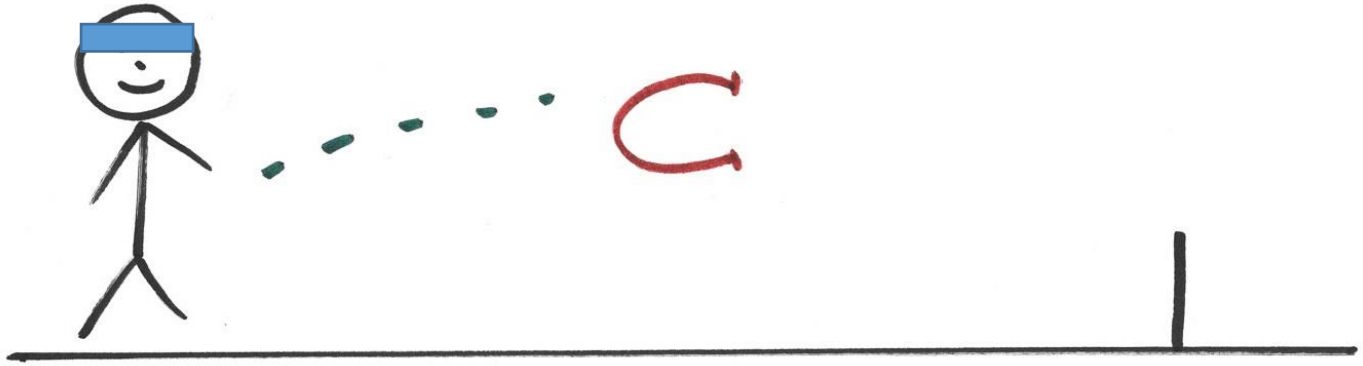
$$150 \pm 8.95 \rightarrow (141.05, 158.95)$$

95% CI for μ: $150 \pm 8.95 \rightarrow (141.05, 158.95)$

We sampled 95% of the population.	false	The confidence level describes how confident we are that the interval captured μ. It has nothing to do with the sample size.
We know that μ = 150 pounds.	false	We don't know the actual value of μ. If we did, then a confidence interval would not be necessary!
95% of MTU students weigh somewhere between 141.05 and 158.95	false	The confidence interval is not trying to capture 95% of the population. It is only trying to capture μ.
We know that $141.05 < \mu < 158.95$	false	Once again, we don't know the true value of μ.
$P(141.05 < \mu < 158.95) = .95$	false	μ is not a random variable, so this statement is meaningless!
We are 95% confident the interval captured μ	true	CORRECT!

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Computing a confidence interval is like playing a game of horseshoes:



horseshoe thrower can't see the pin	horseshoe	pin
researcher doesn't know μ	**confidence interval**	μ

In the long run, 95% of the confidence intervals successfully capture μ, but we aren't completely certain about any particular one.

95% CI for μ

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{30}}\right) \rightarrow 150 \pm 8.95 \rightarrow (141.05,158.95)$$

What if the error margin of 8.95 pounds is considered too large, and we want an error margin of 3 pounds at 95% confidence?

$$Z_{\frac{\alpha}{2}}\ \frac{\sigma}{\sqrt{n}} = B \quad \rightarrow \quad n = \left(\frac{Z_{\frac{\alpha}{2}}\ \sigma}{B}\right)^2$$

$$n = \left(\frac{Z_{\frac{\alpha}{2}}\ \sigma}{B}\right)^2 = \left(\frac{1.96 \cdot 25}{3}\right)^2 \approx 266.78\ \uparrow 267$$

95% confidence interval for μ when σ is known

$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

point estimate

error margin

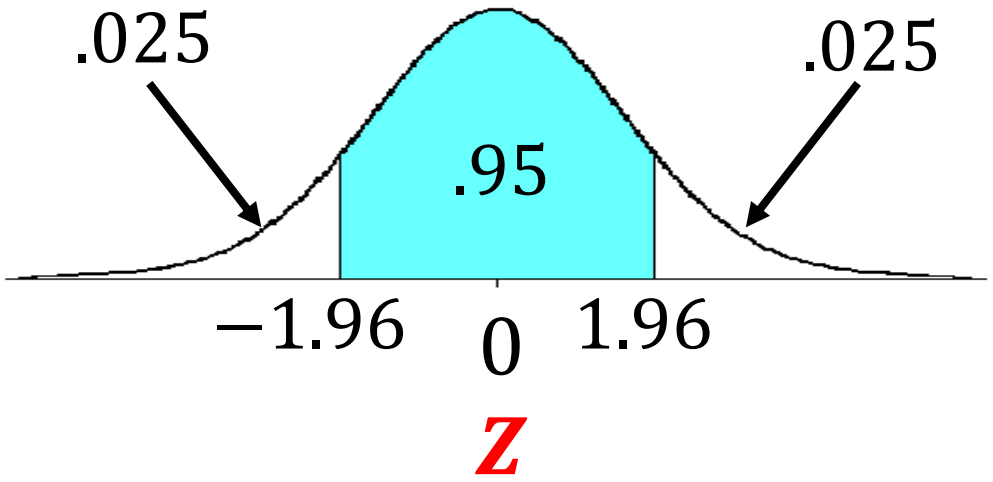
Next section:

What about other confidence levels??

MORE CONFIDENCE INTERVALS FOR μ WHEN σ IS KNOWN

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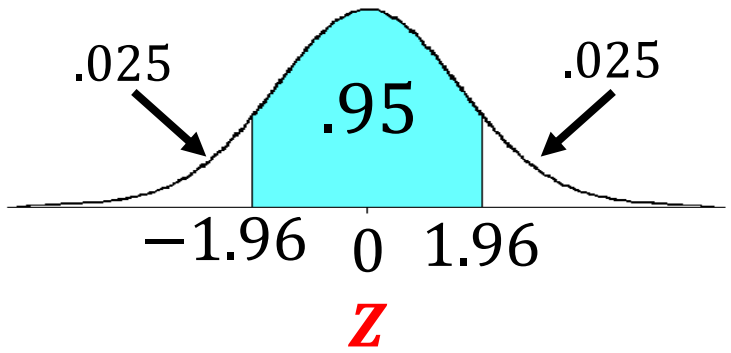
In the previous section we developed a 95% confidence interval for μ when σ is known: $\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$



So if we want a different confidence level, we should replace 1.96 with a different value from the Z curve.

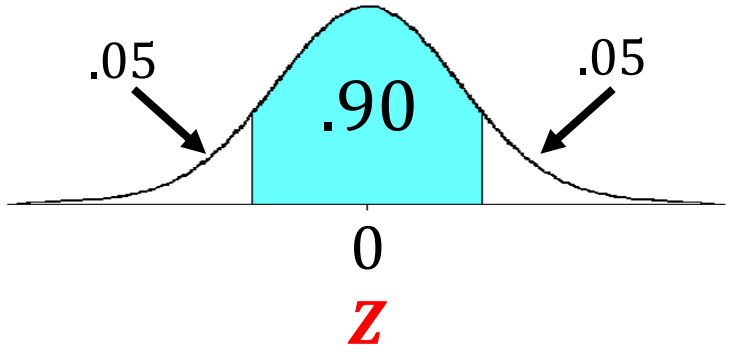
95% confidence

$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .05$ $\frac{\alpha}{2} = .025$

90% confidence



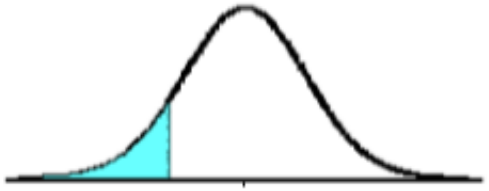
$\alpha = .10$ $\frac{\alpha}{2} = .05$

99% confidence

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Z-Table

negative Z values

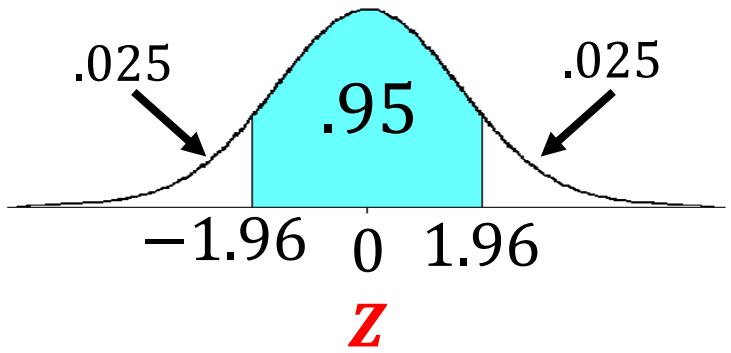


	0.00	0.01	0.02	0.03	0.04	0.05	0.06	
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0

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95% confidence

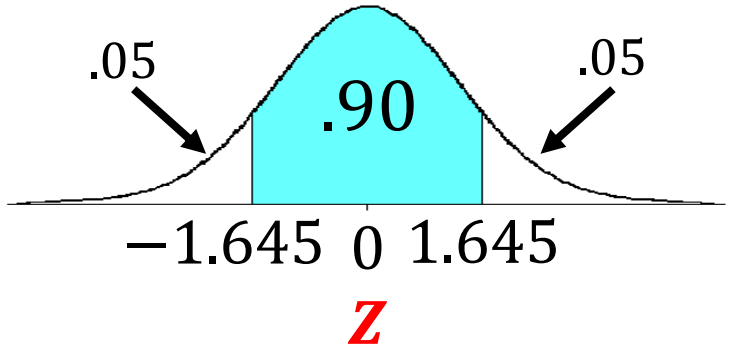
$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .05 \quad \frac{\alpha}{2} = .025$

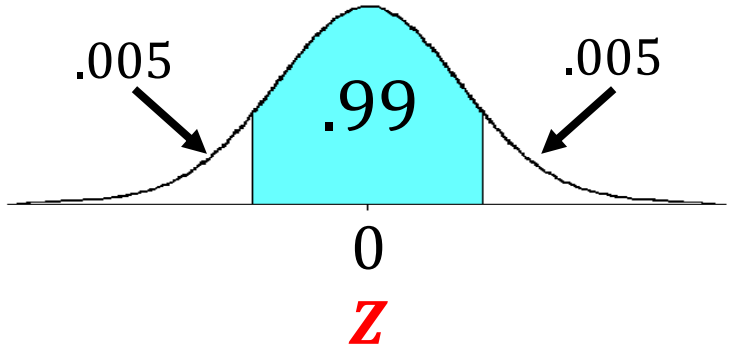
90% confidence

$$\overline{X} \pm 1.645 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .10 \quad \frac{\alpha}{2} = .05$

99% confidence

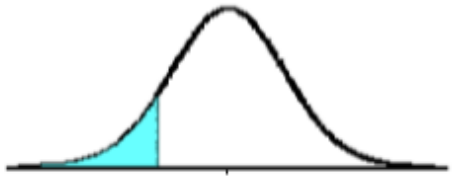


$\alpha = .01 \quad \frac{\alpha}{2} = .005$

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Z-Table

negative Z values

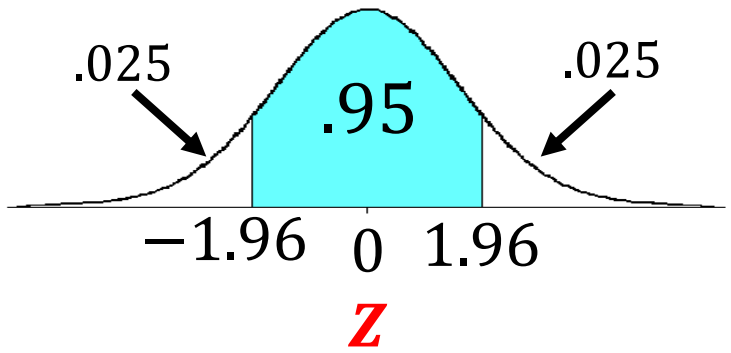


	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.

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95% confidence

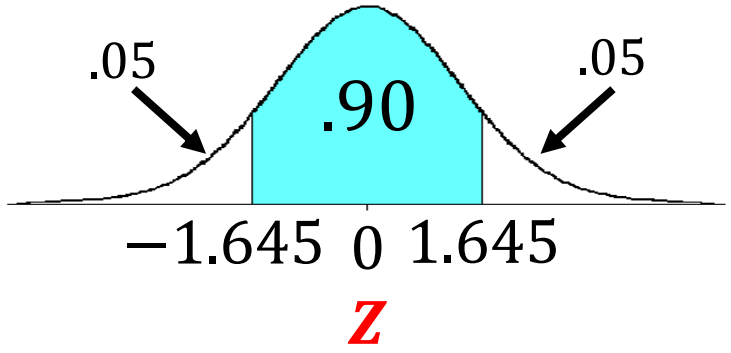
$$\overline{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .05$ $\frac{\alpha}{2} = .025$

90% confidence

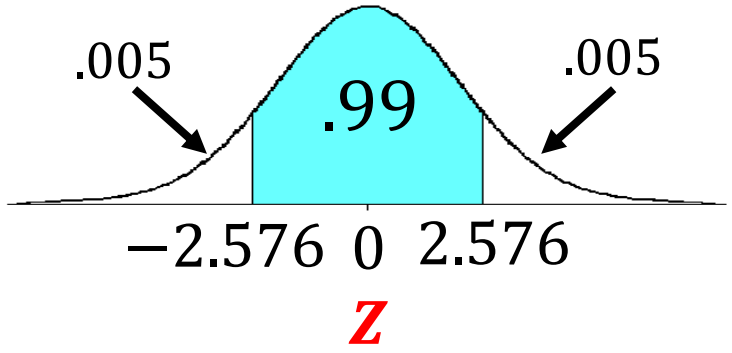
$$\overline{X} \pm 1.645 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .10$ $\frac{\alpha}{2} = .05$

99% confidence

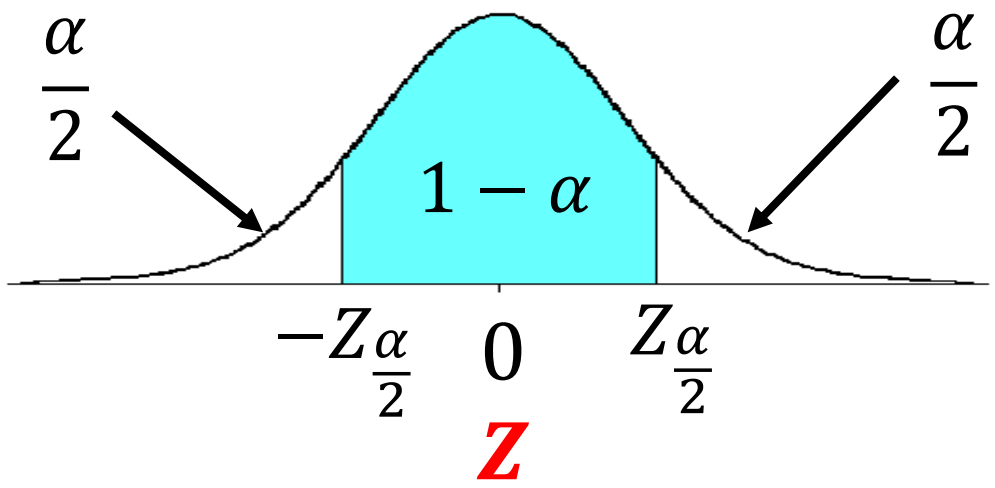
$$\overline{X} \pm 2.576 \frac{\sigma}{\sqrt{n}}$$



$\alpha = .01$ $\frac{\alpha}{2} = .005$

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A $(1-\alpha)100\%$ confidence interval for μ when σ is known (population is normally distributed or $n \geq 30$)



$$\overline{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

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From the previous video:

Suppose we are interested in μ = the population mean weight among all Michigan Tech students.

$n = 30$

$\overline{X} = 150$ pounds

$\sigma = 25$ (known)

95% confidence interval

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{30}}\right)$$

$$150 \pm 8.95 \rightarrow (141.05, 158.95)$$

All other things equal, do we prefer thinner intervals or wider intervals?

(145,155) or (100,400)?

Thinner CI's are better.

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Thinner CI's are better.

What makes a confidence interval wider or thinner?

All else the same, what if the confidence level had been 99%?

95% CI

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{30}}\right)$$

$$150 \pm 8.95 \rightarrow (141.05, 158.95)$$

99% CI

$$\overline{X} \pm 2.576\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 2.576\left(\frac{25}{\sqrt{30}}\right)$$

$$150 \pm 11.76 \rightarrow (138.24, 161.76)$$

Trade off:

It's nice to have more confidence in your interval, but you have to settle for a wider interval.

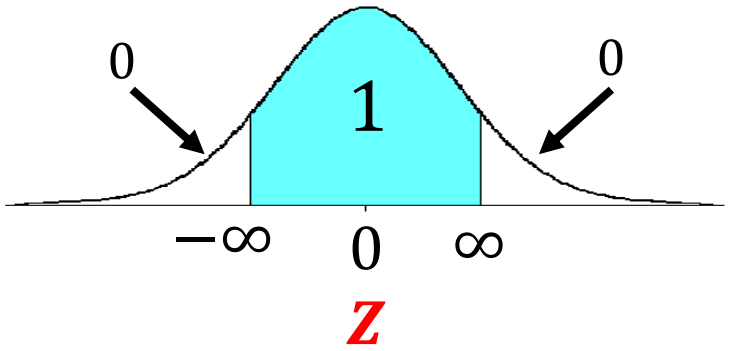
Trade off:

It's nice to have more confidence in your interval, but you have to settle for a wider interval.

What if someone insisted on a completely risk-free confidence interval and demanded a confidence level of 100%?

100% CI

$$\overline{X} \pm \infty \frac{\sigma}{\sqrt{n}} \rightarrow (-\infty, \infty)$$



Although the 100% CI is indeed guaranteed to capture μ, it is completely useless. Lesson: there is no such thing as a risk-free confidence interval (or hypothesis test)!

Thinner CI's are better.

What makes a confidence interval wider or thinner?

All else the same, what if the sample size had been $n = 100$?

95% CI with n = 30

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{30}}\right)$$

$$150 \pm 8.95 \rightarrow (141.05, 158.95)$$

95% CI with n = 100

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}} \rightarrow 150 \pm 1.96\left(\frac{25}{\sqrt{100}}\right)$$

$$150 \pm 4.9 \rightarrow (145.1, 154.9)$$

Larger sample sizes are good. They enable you to have a thinner interval.

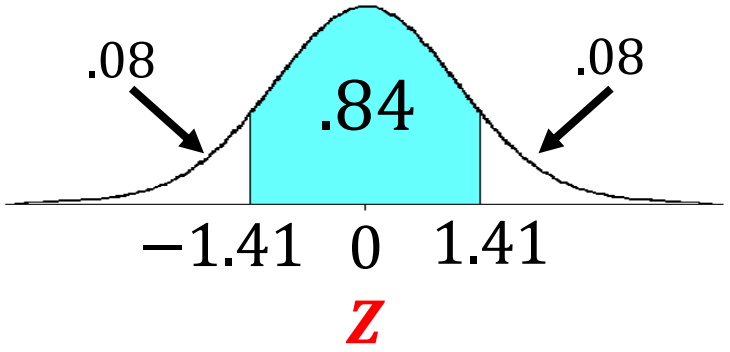
μ = population mean wall thickness (mm) among steel canisters

$n = 70$

$\overline{X} = 8.1$ mm

$\sigma = 0.5$ (known)

Large sample size, therefore we are NOT required to assume the population of wall thicknesses are normally distributed. (CLT Part 2)



Calculate a 84% confidence interval for μ

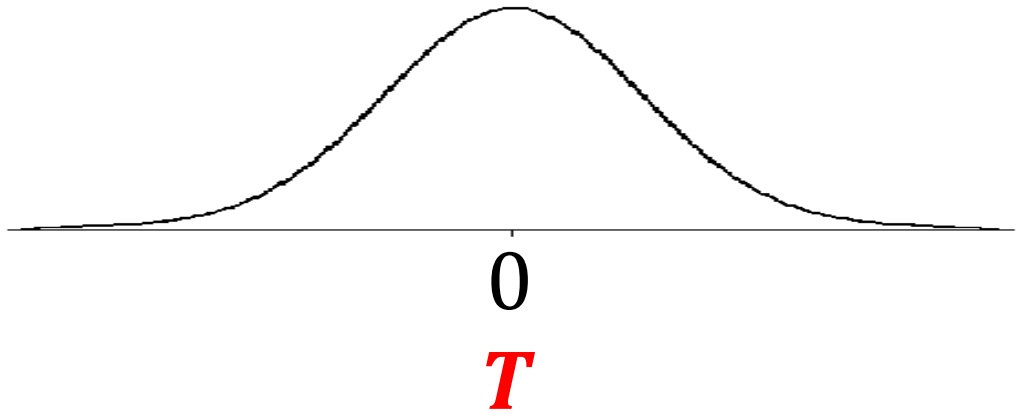
$$\overline{X} \pm 1.41\frac{\sigma}{\sqrt{n}} \rightarrow 8.1 \pm 1.41\left(\frac{0.5}{\sqrt{70}}\right)$$

$$8.1 \pm 0.084 \rightarrow (8.016, 8.184)$$

T CURVE AND T TABLE

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The **T distribution** (or T curve) is a symmetric bell-shaped continuous random variable centered around zero, just like the Z curve, however it is shorter and wider.



The T curve is symmetric around zero.

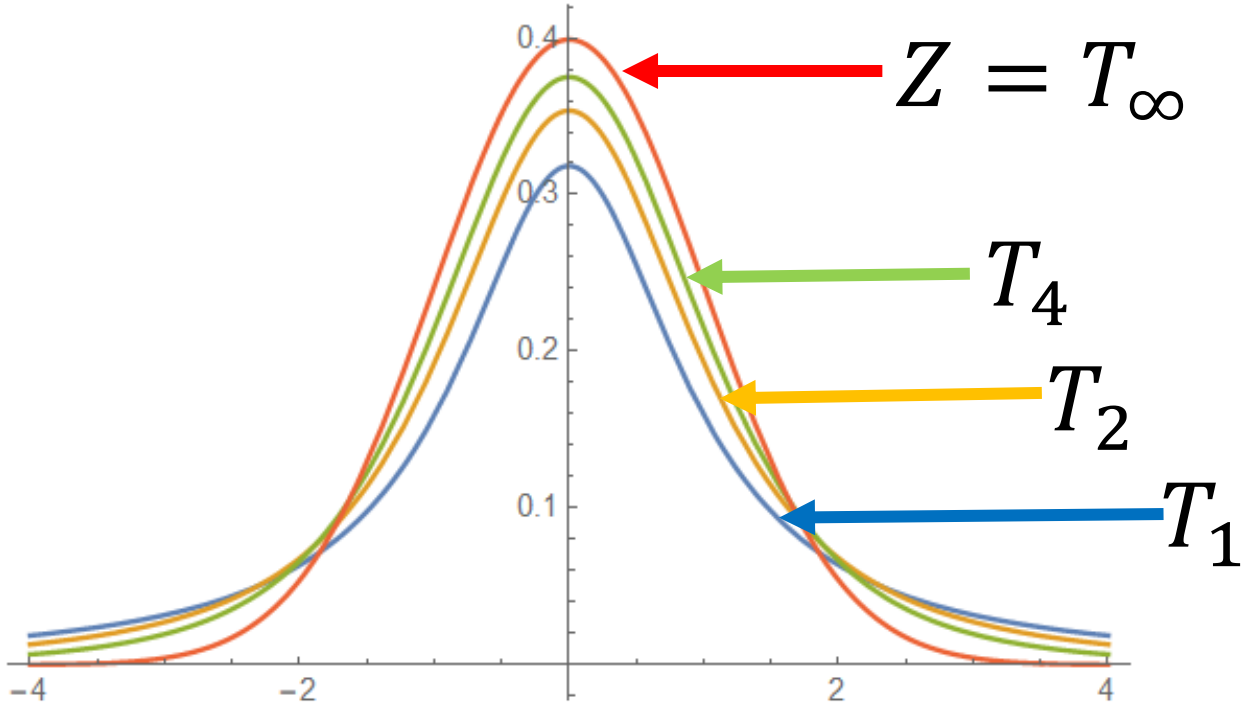
The total area under the T curve is one.

The T curve is shorter and wider than the Z curve.

There are actually an endless number of T curves. They are identified by a number called the degrees of freedom.

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There are an endless number of T curves. They are identified by a number called the degrees of freedom.



In the middle of the T table you will see t-values with a certain area to the RIGHT.

T-Table



	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	0.868	1.345	1.761	2.145	2.624	2.977	3.787	4.140	4.985
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791

area to the right

degrees of freedom

CASE WESTERN RESERVE UNIVERSITY

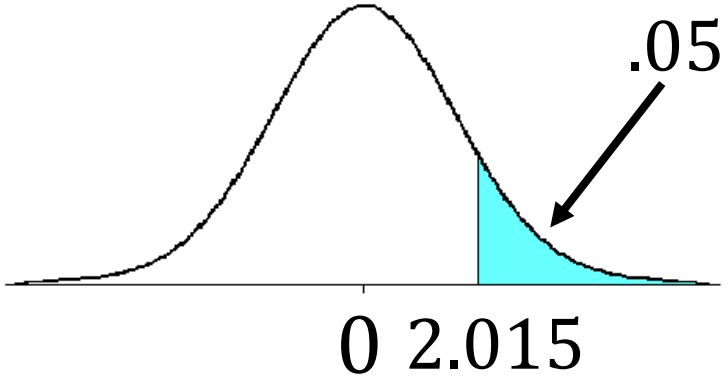
For example, in row 5 and the 0.05 column you will see the t-value of 2.015.

T-Table

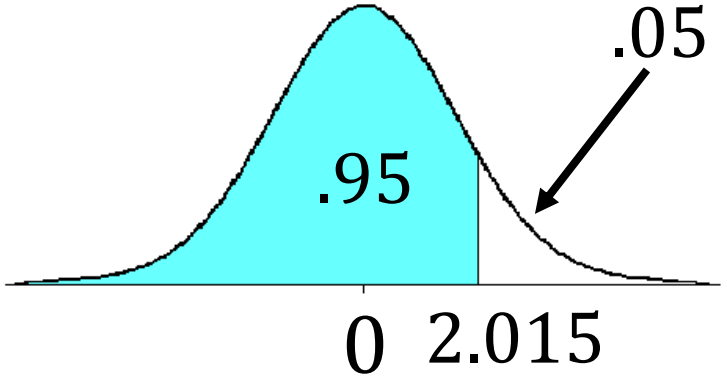


	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
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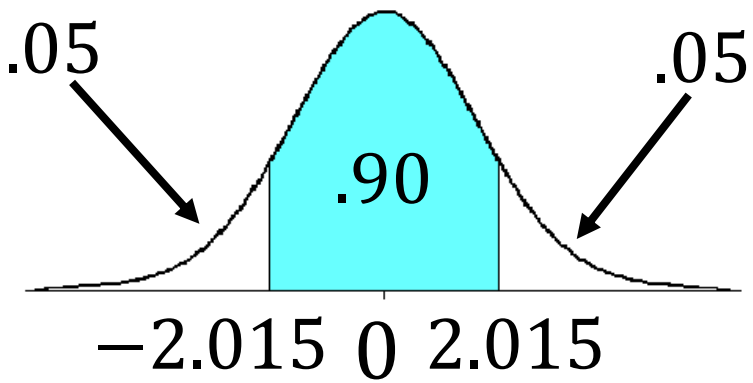
CASE WESTERN RESERVE UNIVERSITY



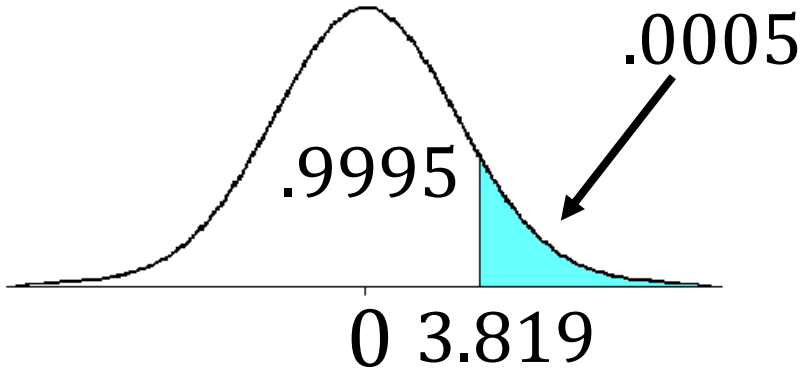
T_5



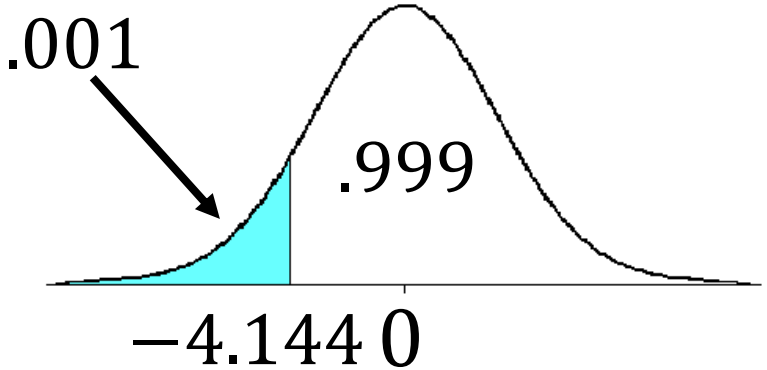
T_5



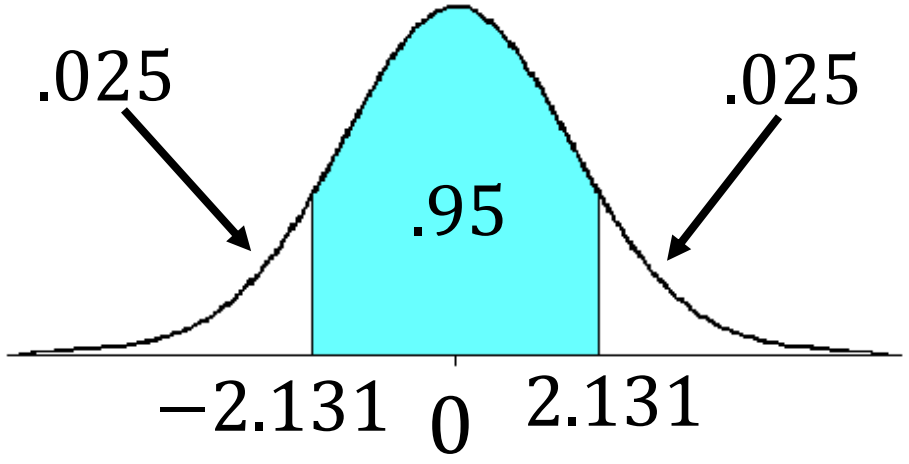
T_5



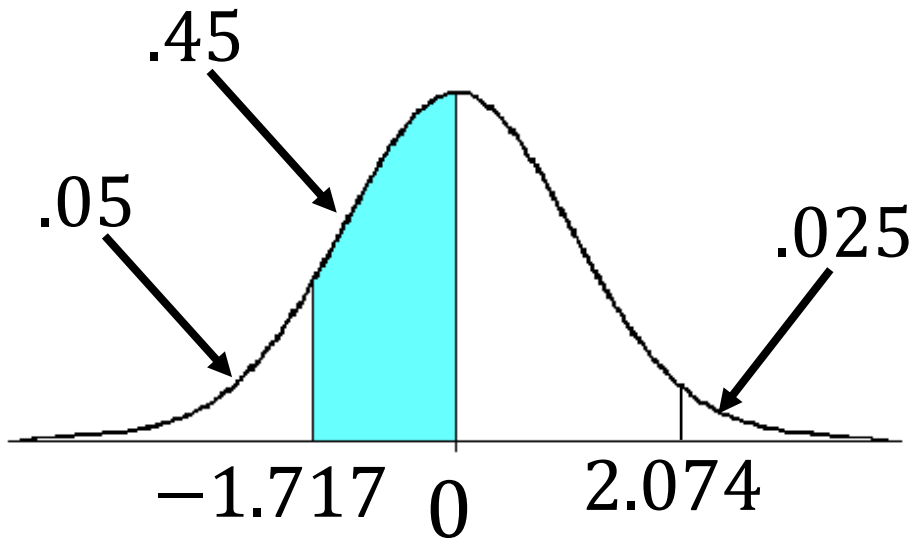
T_{21}



T_{10}



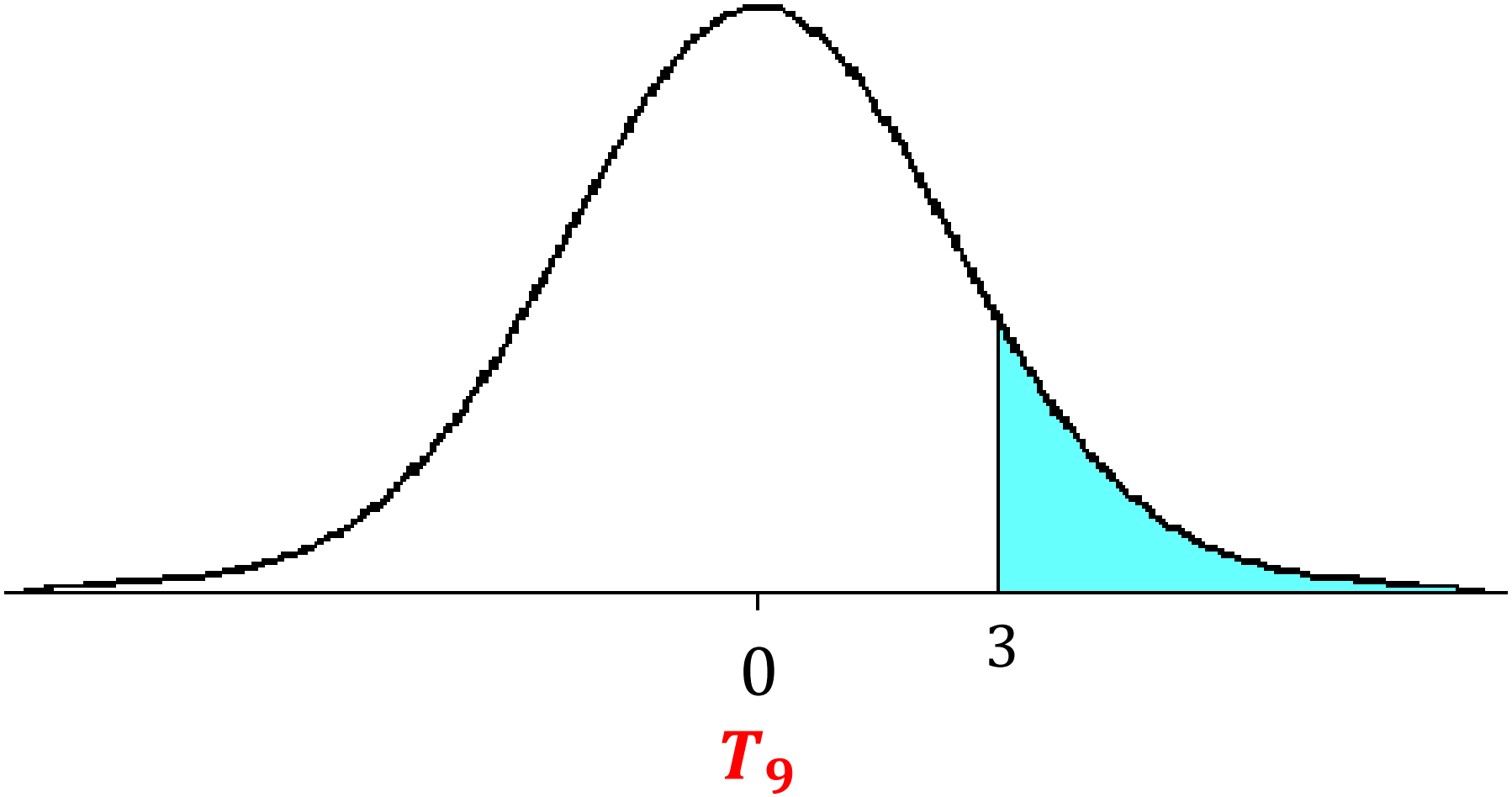
T_{15}



T_{22}

CASE WESTERN RESERVE UNIVERSITY

Place bounds on the shaded area.



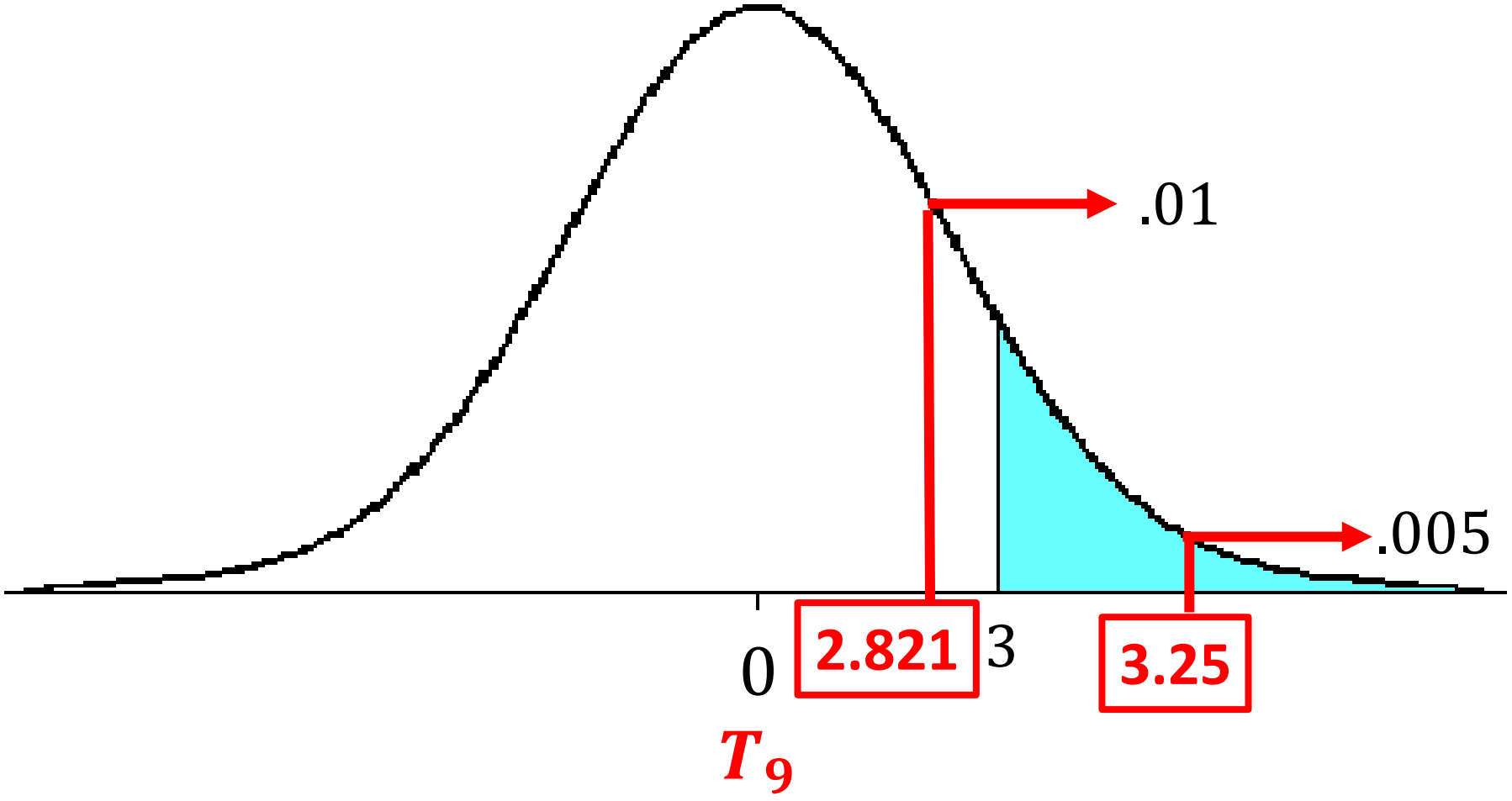
Look for two values in row 9. One value should be slightly less than 3, the other value should be slightly greater than 3.

T-Table



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1	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
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CASE WESTERN RESERVE UNIVERSITY

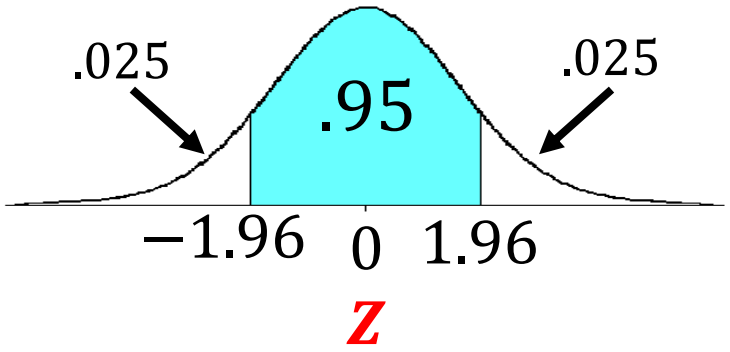


.005 < shaded area < .01

Notice that as the degrees of freedom increase, the t-values decrease, approaching the corresponding z-value.

.025 column

25	0.856	1.316	1.708	2.060	2.485	2.787	3.450	3.725	4.352
26	0.856	1.315	1.706	2.056	2.479	2.779	3.435	3.707	4.324
27	0.855	1.314	1.703	2.052	2.473	2.771	3.421	3.690	4.299
28	0.855	1.313	1.701	2.048	2.467	2.763	3.408	3.674	4.275
29	0.854	1.311	1.699	2.045	2.462	2.756	3.396	3.659	4.254
30	0.854	1.310	1.697	2.042	2.457	2.750	3.385	3.646	4.234
40	0.851	1.303	1.684	2.021	2.423	2.704	3.307	3.551	4.094
50	0.849	1.299	1.676	2.009	2.403	2.678	3.261	3.496	4.014
60	0.848	1.296	1.671	2.000	2.390	2.660	3.232	3.460	3.962
70	0.847	1.294	1.667	1.994	2.381	2.648	3.211	3.435	3.926
80	0.846	1.292	1.664	1.990	2.374	2.639	3.195	3.416	3.899
90	0.846	1.291	1.662	1.987	2.368	2.632	3.183	3.402	3.878
100	0.845	1.290	1.660	1.984	2.364	2.626	3.174	3.390	3.862
200	0.843	1.286	1.653	1.972	2.345	2.601	3.131	3.340	3.789
500	0.842	1.283	1.648	1.965	2.334	2.586	3.107	3.310	3.747
∞	0.842	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.719

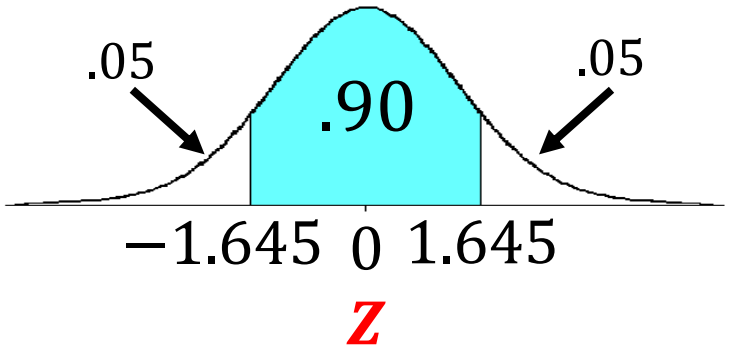


CASE WESTERN RESERVE UNIVERSITY

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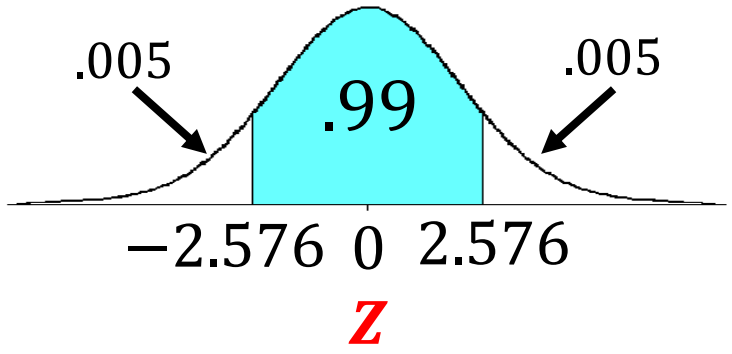


CASE WESTERN RESERVE UNIVERSITY

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.005 column

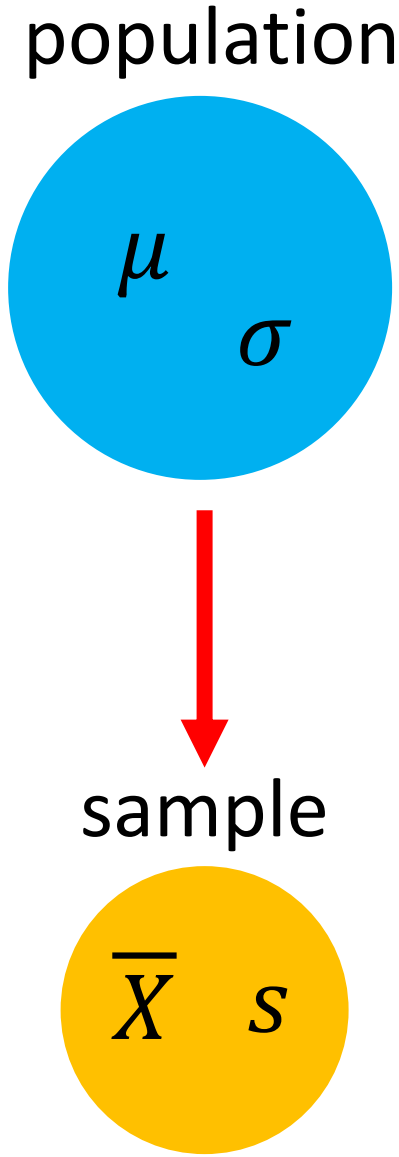
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CASE WESTERN RESERVE UNIVERSITY

CONFIDENCE INTERVAL FOR μ WHEN σ IS UNKNOWN

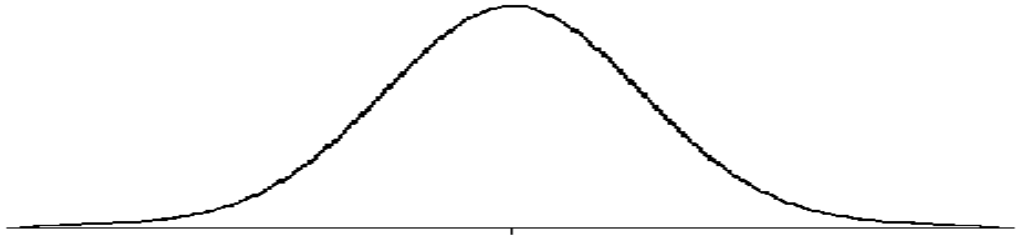
CASE WESTERN RESERVE UNIVERSITY EST. 1826
think beyond the possible℠



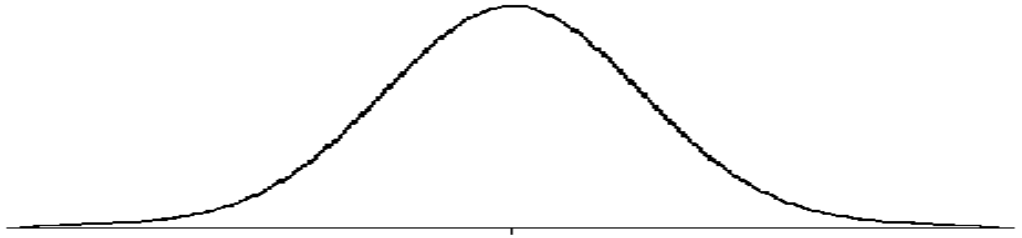
So far, we have assumed that σ is known.

But if μ is unknown, then σ would most unlikely be unknown as well. Now we will consider this more realistic situation.

In addition to estimating μ with $\overline{X}$, we will estimate σ with s.



Z



T

Before:

σ known

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$

Now:

σ unknown

$$T = \frac{\overline{X} - \mu}{s / \sqrt{n}}$$

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A $(1-\alpha)100\%$ confidence interval for μ when **σ is known** (population is normally distributed or $n \geq 30$)

$$\overline{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

A $(1-\alpha)100\%$ confidence interval for μ when **σ is unknown** (population is normally distributed or $n \geq 30$)

$$\overline{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

Important!

When doing a one-sample confidence interval or hypothesis test for μ:

Use the Z curve if **σ is known**

Use the T curve if **σ is unknown**

Our policy in this course:

When doing a one-sample confidence interval or hypothesis test for μ:

Use the Z curve if **σ is known**

Use the T curve if **σ is unknown**

Warning

Many courses/textbooks use this slightly different policy.

(Technically, this is incorrect.)

	$n \geq 30$	$n < 30$
σ known	**Z**	**Z**
σ unknown	**Z (close enough)**	**T**

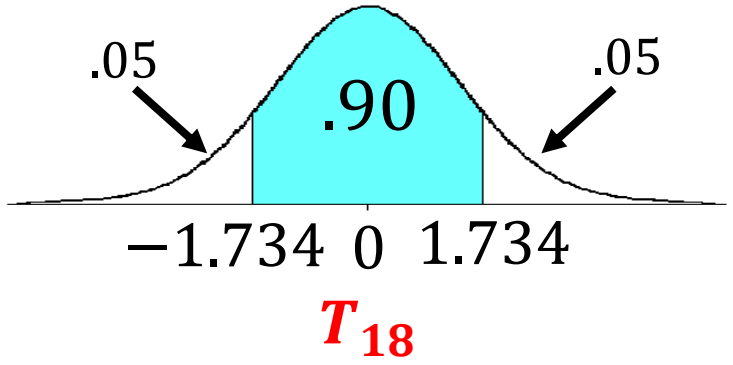
μ = population mean cholesterol level (mg/dL) among 70-year-old American men

$n = 19$

$\overline{X} = 44.2$ mg/dL

$s = 3.8$ (σ unknown)

Small sample size, therefore we must assume the population of cholesterol levels are normally distributed and we should examine a QQ plot. (CLT Part 1)



Calculate a 90% confidence interval for μ

$$\overline{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \rightarrow 44.2 \pm 1.734 \left(\frac{3.8}{\sqrt{19}}\right)$$

$$44.2 \pm 1.51 \rightarrow (42.69, 45.71)$$

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90% confidence interval for $\mu \rightarrow (42.69, 45.71)$

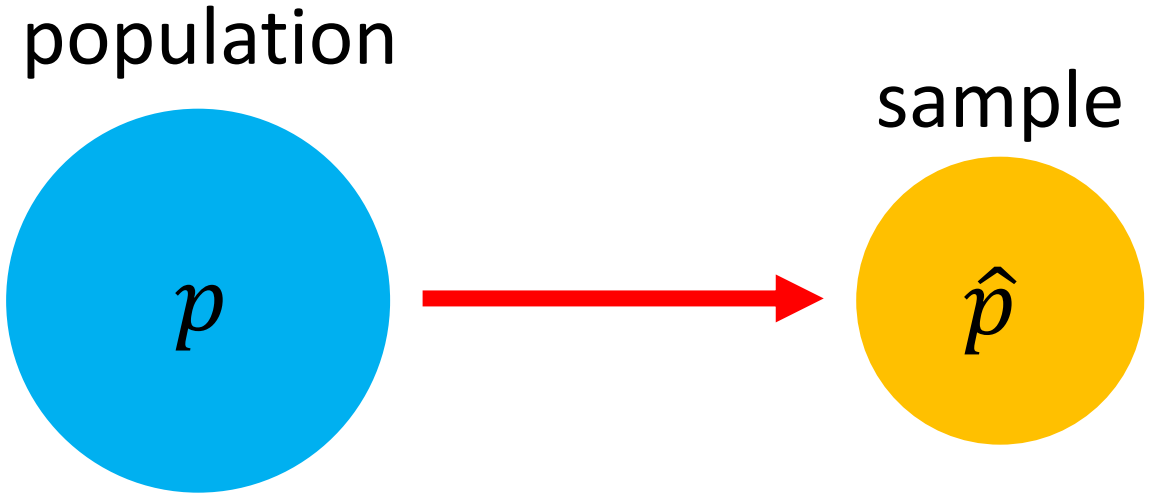
Did we sample 90% of the population?	no
Do 90% of 70-year-old American men have a cholesterol level between 42.69 and 45.71 mg/dL?	no
Are we 90% confident the population mean cholesterol level among 70-year-old American men has been captured by the interval?	yes

CONFIDENCE INTERVAL FOR P

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think beyond the possible℠

Recall:

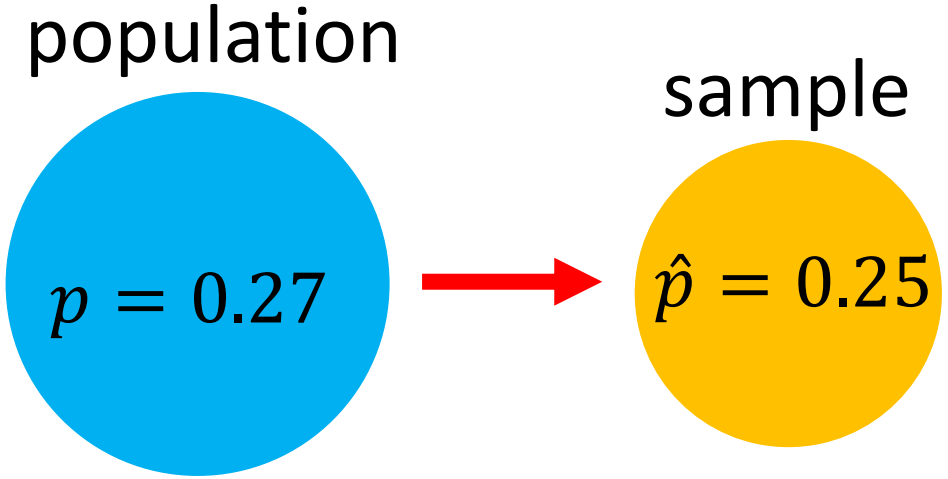
A proportion is simply a percentage expressed as a fraction or decimal between zero and one.



The sample proportion $\hat{p}$ is an estimator of the unknown population proportion p.

Example:

$p = 0.27$ is the proportion of students who walk to school every day. A researcher who doesn't know this will estimate p by asking 100 randomly selected students if they walk to school. 25 of these students say they walk to school.



$$\hat{p} = \frac{X}{n} = \frac{25}{100} = 0.25$$

X = the # of people or things in the sample with the characteristic of interest

n = sample size

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X = the # of people or things in the sample with the characteristic of interest

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Is X a random variable? **Yes**

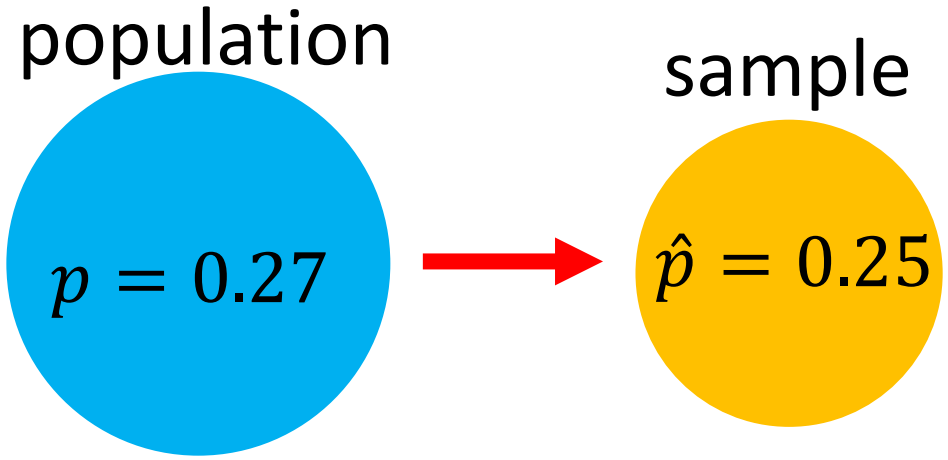
Is X a discrete or continuous random variable? **discrete**

What are the possible values of X?

$X = \{0, 1, 2, \ldots, 100\}$ or more generally $\boldsymbol{X = \{0, 1, 2, \ldots, n\}}$

Is X one of the famous discrete random variables we've studied?

Yes! $\boldsymbol{X \sim BIN(n, p)}$



$$\hat{p} = \frac{X}{n} = \frac{25}{100} = 0.25$$

Recall:

If $X \sim BIN(n, p)$

X counts the # of successes in n trials.

p = P(success). This is the same p.

$\mu = E(X) = np$. In this example, $\mu = E(X) = 100(.27) = 27$

$$\sigma^2 = Var(X) = np(1-p)$$

In this example, $\sigma^2 = Var(X) = 100(.27)(.73) = 19.71$

$\sigma = SD(X) = \sqrt{\sigma^2}$. In this example, $\sigma = \sqrt{19.71} \approx 4.44$

When we first developed a confidence interval for μ we went through a 3-item checklist:

1

Is the point estimate normal or approximately normal?

Yes, if the population is normally distributed (CLT 1) or the sample size is large (CLT 2)

When we first developed a confidence interval for μ we went through a 3-item checklist:

2

Is the point estimate unbiased?

Yes, because $E(\overline{X}) = \mu$

3

What is the standard deviation of the point estimate?

$$SD(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\overline{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

Now, let us go through the same 3-item checklist as we develop a confidence interval for p:

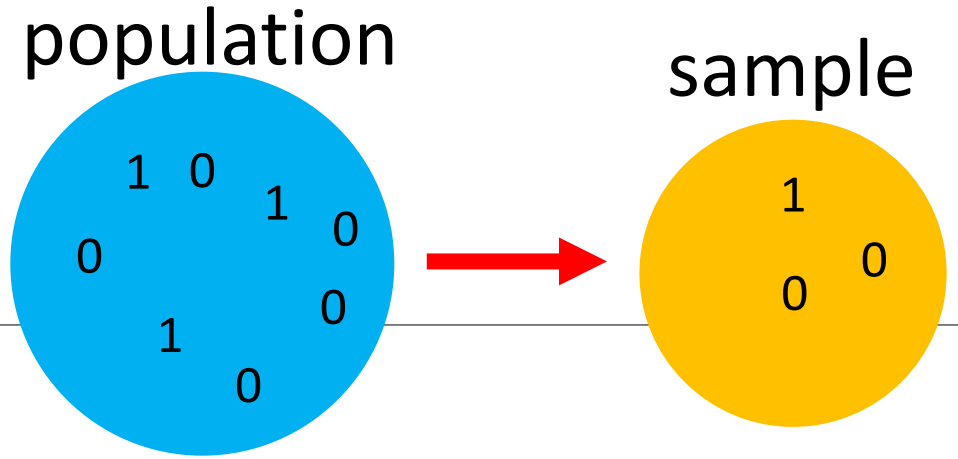
1

Is the point estimate normal or approximately normal?

A population of 1's and 0's can't be normally distributed.

Therefore we must use Central Limit Theorem Part 2 and have a large sample size.

<u>All proportion problems must have large sample sizes</u>!



Now, let us go through the same 3-item checklist as we develop a confidence interval for p:

2

Is the point estimate unbiased?

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}(np) = p$$

Yes! $E(\hat{p}) = p$

Now, let us go through the same 3-item checklist as we develop a confidence interval for p:

3

What is the standard deviation of the point estimate?

$$Var(\hat{p}) = Var\left(\frac{X}{n}\right) = \frac{1}{n^2}Var(X) = \frac{1}{n^2}\big(np(1-p)\big) = \frac{p(1-p)}{n}$$

$$SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

All Z and T confidence intervals follow a basic recipe:

Point Estimate $\pm$ (Z or T) $\cdot$ SD(Estimator)

$$\overline{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \qquad \overline{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

All Z and T confidence intervals follow a basic recipe:

Point Estimate $\pm$ (Z or T) $\cdot$ SD(Estimator)

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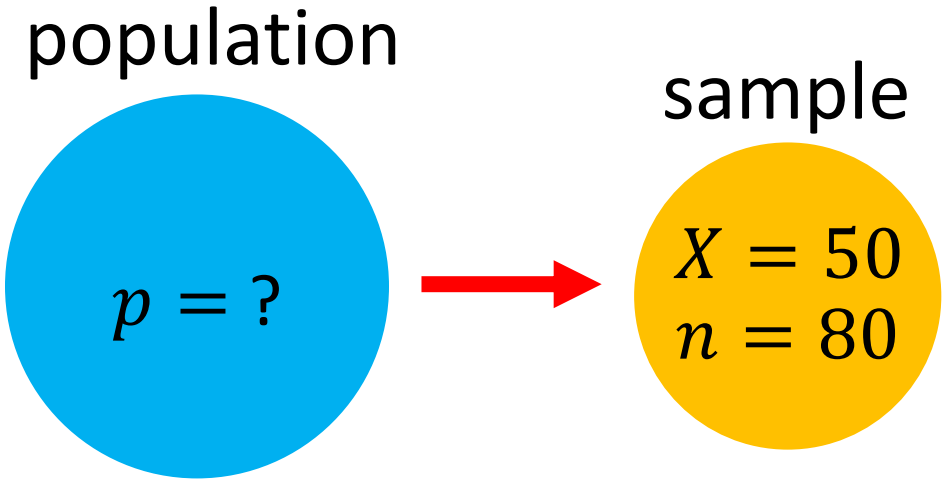
A $(1-\alpha)100\%$ confidence interval for p $(n \geq 30)$

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

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Example:

p =the proportion of female basketball players who are over 68 inches tall.



$$\hat{p} = \frac{X}{n} = \frac{50}{80} = 0.625$$

Compute a 95% confidence interval for p

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Example:

p =the proportion of female basketball players who are over 68 inches tall.

$$\hat{p} = \frac{X}{n} = \frac{50}{80} = 0.625$$ Compute a 95% confidence interval for p

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \rightarrow 0.625 \pm 1.96 \sqrt{\frac{0.625(0.375)}{80}}$$

$$0.625 \pm 0.106 \rightarrow (0.519, 0.731)$$

We are 95% confident the proportion of female basketball players taller than 68 inches is between 0.519 and 0.731. (51.9% and 73.1%)

$$0.625 \pm 0.106 \rightarrow (0.519, 0.731)$$

We are 95% confident the proportion of female basketball players tall than 69 inches is between 0.519 and 0.731. (51.9% and 73.1%)

What if the error margin of 0.106 is considered too large and a smaller error margin of 0.04 is desired while maintaining the 95% confidence level? What sample size is necessary?

$$Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = B \quad \rightarrow \quad n = \hat{p}(1-\hat{p}) \left(\frac{Z_{\frac{\alpha}{2}}}{B} \right)^2$$

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$$Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = B \quad \rightarrow \quad n = \hat{p}(1-\hat{p})\left(\frac{Z_{\frac{\alpha}{2}}}{B}\right)^2$$

$$n = \hat{p}(1-\hat{p})\left(\frac{Z_{\frac{\alpha}{2}}}{B}\right)^2 = 0.625(0.375)\left(\frac{1.96}{0.04}\right)^2 \approx 562.7 \uparrow 563$$

The sample size formula $n = \hat{p}(1-\hat{p})\left(\frac{Z_{\frac{\alpha}{2}}}{B}\right)^2$ requires a prior estimate of p. If you do not have a $\hat{p}$, then use $\hat{p} = .50$. (This maximizes the answer.)

For example:

Before collecting any data someone knows they would like their 90% confidence interval for p to have an error margin of 0.02. What sample size is necessary?

$$n = \hat{p}(1-\hat{p})\left(\frac{Z_{\frac{\alpha}{2}}}{B}\right)^2 = 0.5(0.5)\left(\frac{1.645}{0.02}\right)^2 \approx 1691.3 \uparrow 1692$$